SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Error Analysis Task

Course Code:	CSA17-16	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 3 – Error Analysis Task
Weightage:	35% of CIA	**Duration / Marks:**	60 mins / 100 marks
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
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Section / Batch:		**Date:**	10 October 2026

M. SRILEKHA / 1919260007

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

M. SRILEKHA

Instructions

- Read each question carefully; answer all four questions (100 in total). Off-Class Assessment Tool.
- Each question gives a student's faulty work. Do not solve from scratch: identify every error (cite it by statement, component, iteration or step number), explain its root cause, correct it with justification and apply the corrected concept. An error merely carried forward counts once.
- Follow the conventions stated in each question (alphabetical expansion, explored set, goal test on removal from the frontier, tie-breaking) and present tables neatly.
- Marks follow the Error Analysis rubric: error identification, root cause analysis, correction and justification, application of concepts, and clarity and presentation.
- Upload in GitHub and GCR (question paper with Reg. No., Name, answers, rubrics and evaluation page at the end).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Four approaches to AI, Turing Test, eras and milestones	**Q1**
2. Intelligent Agents: Agents and Environments	Agent, percepts, actions, agent function and program, PEAS	**Q1**
3. The Concept of Rationality	Performance measure, percept sequence, rational vs omniscient	**Q1**
4. The Nature of Environments	Task-environment properties and their classification	**Q1**
5. The Structure of Agents	Simple reflex and model-based reflex agent programs	**Q1**
6. Problem-Solving Agents	Goal formulation, problem formulation, search, execution	**Q2**
7. Problem Formulation	States, initial state, actions, goal test, path cost; belief states	**Q2, Q4**
8. Searching for Solutions	Search graph, frontier, explored set, node expansion	**Q3**

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QUESTION 1: Foundations of AI, Intelligent Agents and Rationality

(a) Identify the Correct and Incorrect Statements

Statement Correct / IncorrectErroneous phrase / claim

S1 Incorrect TheTuring Test does not check whether internal reasoning always follows logic. It tests intelligent behaviour through conversation.
S2 Incorrect AI winters did not occur in the 1950s before the Dartmouth workshop. AI winters occurred later, during periods of reduced funding and interest.
S3 Incorrect A rational agent does not know every action's result in advance. It makes decisions using available information and expected outcomes.
S4 Incorrect Performance measure and actuators are interchanged. Safe, fast, legal and comfortable trips are performance measures. Steering, accelerator, brake and horn are actuators.
S5 Incorrect Chess without a clock is fully observable, deterministic, sequential, static and discrete.
S6 CorrectA model-based reflex agent maintains an internal state and uses a model of how the world changes.

(b) Root Cause Analysis

Error Root cause (misconception)

S1 The student confused the Turing Test with the thinking-rationally approach. The Turing Test evaluates intelligent behaviour rather than the logical correctness of internal reasoning.
S2 The student misunderstood the history of AI. The Dartmouth workshop took place in 1956, while AI winters occurred later.
S3 The student confused rationality with omniscience. A rational agent uses available information to choose the best expected action but cannot predict every future event.
S4 The student confused the components of PEAS. Performance measures describe success, while actuators perform actions.
S5 The student misunderstood task-environment properties. Chess has complete information, predictable rules, sequential decisions, a static board and discrete states and actions.
S6 No error. The statement correctly explains the internal state and world model of a model-based reflex agent.

(c) Correct the Statements and Give Justification

S1

Corrected statement:

The Turing Test, proposed by Alan Turing in 1950, evaluates whether a machine can exhibit behaviour that is indistinguishable from a human in a text-based conversation. It is an example of the acting-humanly approach to AI.

Justification: The Turing Test evaluates observable behaviour, not whether internal reasoning always follows logical rules.

S2

Corrected statement:

The term Artificial Intelligence was introduced at the Dartmouth workshop in 1956. AI winters occurred later, during periods when funding and interest in AI research declined.

Justification: The Dartmouth workshop is considered an important milestone in the establishment of AI as a research field. AI winters occurred in later periods.

S3

Corrected statement:

A rational agent selects the action expected to maximise its performance measure based on its percept sequence, available knowledge and possible actions. It does not need to know the actual outcome of every action in advance.

Justification: Rationality depends on the information available when the decision is made. An unforeseeable landslide does not automatically make the taxi's earlier decision irrational.

S4

Corrected statement:

The PEAS description of an automated taxi is:

Component	Correct description
Performance measure	Safe, fast, legal and comfortable trip
Environment	Roads, other traffic, pedestrians and weather

Actuators Steering, accelerator, brake and horn
SensorsCameras, GPS and speedometer

Justification: PEAS stands for Performance measure, Environment, Actuators and Sensors. Each component has a different role in describing an intelligent agent's task.

S5

Corrected statement:

Chess played without a clock is a fully observable, deterministic, sequential, static and discrete task environment.

Justification:

Fully observable: Both players can see all the pieces on the board.

Deterministic: The result of a legal move is predictable.

Sequential: Each move affects future decisions.

Static: The board does not change while a player is thinking.

Discrete: The board positions, moves and states are countable.

S6

Corrected statement:

No change.

A model-based reflex agent maintains an internal state, updated using knowledge of how the world evolves and how its own actions affect it. It can act sensibly even when the current percept does not reveal the full state of the world.

Justification: The statement correctly describes how an internal state and a model help an agent make decisions when information is incomplete.

(d) Apply the Corrected Concepts

(i) Was the taxi's choice rational?

The taxi's choice was rational because all available traffic reports showed that the

highway was clear when it selected the route.

The four factors that determine rationality are:

1. Performance measure: The taxi must provide a safe, fast and comfortable journey.

2. Percept sequence: The available traffic reports indicated that the highway was clear.

3. Knowledge of the environment: The taxi used the information available about the road and traffic conditions.

4. Available actions: The taxi selected the highway route from the available routes.

The landslide was unforeseeable. Therefore, the taxi's original decision can still be rational even though the journey was interrupted.

(ii) Name the task-environment properties that make perfect foresight impossible.

The relevant properties are:

Partially observable: The taxi's sensors cannot detect every condition in the environment.

Stochastic: Unexpected events, such as landslides, may occur.

Dynamic: Road and traffic conditions can change while the taxi is travelling.

These properties prevent the taxi from knowing every future event with certainty.

(iii) Give one situation where a simple reflex taxi behaves poorly.

Situation: A simple reflex taxi sees a red traffic light and stops. When the light turns green, it starts moving without considering that a pedestrian is still crossing the road.

Problem: A simple reflex agent acts only on the current percept and may not remember relevant earlier information.

Model-based reflex taxi: It maintains an internal state containing relevant information about the environment and uses a model of how the environment changes. This helps it make better decisions when the current percept alone is insu ffi cient.

QUESTION 2: Problem-Solving Agents and Problem Formulation

(a) Identify Correct and Incorrect Components

Component / Note Correct / Incorrect What is wrong?

State representation Incorrect The state must include the boat's position along with the numbers of missionaries and cannibals.
Initial state Incorrect The initial state should be (3, 3, Left), including the boat's position.
Actions Correct The listed actions represent all possible boat loads of one or two people.
Goal test Incorrect M = C is not the goal test. The goal is to move all missionaries and cannibals to the right bank.
Legality check Incorrect The student checks only the left bank. Both banks must satisfy the safety condition.
Path cost Incorrect The path cost should be the number of boat crossings, not the number of people carried.
Note N1 Incorrect The correct order is goal formulation → problem formulation → search → execution.
Note N2 Incorrect There are 20 possible states in the full representation, not 16. The number of legal states is not 10.

(b) Root Cause Analysis

Error Root cause

State representation The student forgot to include the boat's position, which affects the available actions.
Initial state The initial state is incomplete because it does not specify the boat's position.
Goal test The student confused equal numbers on the left bank with the actual goal of moving everyone to the right bank.
Legality check The student ignored the safety condition on the right bank.
Path cost The student confused the number of people carried with the number of crossings.
Note N1 The student misunderstood the order of problem-solving agent operations.
Note N2 The student incorrectly counted the possible states in the state space.

(c) Correct the Faulty Components and Justify

Component / Note	Corrected formulation	Justification
State representation	(M, C, B), where M = missionaries on the left, C = cannibals on the left, and B = boat position (Left or Right).	The boat's position determines which crossing actions are possible.
Initial state	(3, 3, Left)	Everyone starts on the left bank, and the boat is on the left.
Actions	1M, 2M, 1C, 2C, 1M + 1C	These are the possible boat loads. Only safe resulting states are allowed.
Goal test	M = 0 and C = 0	Everyone must reach the right bank.
Legality check	On each bank, missionaries must not be outnumbered by cannibals whenever missionaries are present.	Both banks must be safe.
Path cost	Each boat crossing costs 1.	The objective is to minimise the total number of crossings.
Note N1	Goal formulation → Problem formulation → Search → Execution.	The agent first defines its goal, formulates the problem, searches for a solution and then executes it.
Note N2	There are 32 possible states in the full representation before applying safety constraints.	There are 4 possible values for M, 4 for C and 2 for boat position: $4 \times 4 \times 2 = 32$.

(d) Apply the Corrected Formulation

(i) Complete the table of boat loads from the initial state

Initial state: (3, 3, Left)

Boat load	Resulting state (M, C, Boat)	Legal?	Reason
1M	(2, 3, Right)	Illegal.	On the left bank, 2 missionaries and 3 cannibals remain.
2M	(1, 3, Right)	Illegal.	On the left bank, 1 missionary and 3 cannibals remain.
1C	(3, 2, Right)	Legal.	Missionaries are not outnumbered on either bank.
2C	(3, 1, Right)	Legal.	Both banks satisfy the safety condition.
1M + 1C	(2, 2, Right)	Legal.	Missionaries are not outnumbered on either bank.

Note: A state is legal if the number of missionaries is zero or is at least equal to the number of cannibals on each bank.

(ii) Give one state accepted by the student's legality check that is actually illegal.

Answer: (1, 3, Right).

The student's check considers only the left bank. Since 3 cannibals outnumber 1

missionary on the left bank, the state is actually illegal.

(iii) State the cost of an optimal solution.

Answer: The optimal solution requires 11 boat crossings, so the total path cost is 11.

Each crossing has a cost of 1. Therefore:

Total path cost = 11 × 1 = 11.

QUESTION 3: Searching for Solutions and Uninformed Search

(a) Identify Every Error

No.	Trace / Iteration / Remark	What is wrong?
1	BFS – Iteration 3	The queue is incorrect because node A should not be added again if it is already in the queue.
2	BFS – Iteration 6	The explored set is missing G after the goal is removed from the frontier.
3	BFS result	The path cost is incorrect. The path S–D–G costs 3 + 6 = 9, but BFS does not guarantee the lowest path cost in a weighted graph.
4	DFS – Iteration 1	The children are pushed in the wrong order. Alphabetically first child C should be popped first.
5	DFS result	DFS does not guarantee an optimal solution.
6	UCS – Iteration 3	The priority queue is incorrect because the path to G through D costs 9, while the route through E may provide a cheaper path.
7	Remark R1, R2 and R3	BFS needs an explored set; all three algorithms are not necessarily optimal; DFS is not automatically the best strategy just because it removes fewer nodes.

(b) Root Cause Analysis

No.	Root cause
1	The student did not follow the rule that nodes already in the queue must not be added again.
2	The student did not update the explored set correctly when the goal node was removed.
3	The student confused finding a path with finding the lowest-cost path.
4	The student misunderstood the stack order in DFS.
5	DFS explores depth-first and does not guarantee the lowest-cost path.
6	The student did not correctly compare path costs and update the priority queue in UCS.
7	The student misunderstood explored sets, optimality and the comparison of search

strategies.

(c) Corrected Traces and Justifications

(i) BFS: Corrected Queue After Iteration 3

The graph connections are:

S–C = 3

S–D = 3

A–B = 2

A–C = 6

B–D = 4

B–G = 3

D–E = 2

D–G = 6

E–G = 3

Queue after iteration 3: [B, E, G, A]

Order of dequeued nodes: S, C, D, B, E, G.

BFS finds the goal at depth 2 through S–D–G.

(ii) DFS: Corrected Trace

Iteration	Node popped	Stack after pushing children
1	S	[C, D]
2	C	[A, D]
3	A	[B, D]
4	B	[G, D]
5	G (goal)	[D]

Explanation: DFS uses a stack. The alphabetically first child must be popped first, so children are pushed in the correct order.

(iii) UCS: Corrected Trace

Iteration	Node expanded	g(n)	Priority queue after expansion
1	S	0	C:3 (S), D:3 (S)
2	C	3	D:3 (S), A:9 (C)
3	D	3	E:5 (D), B:7 (D), G:9 (D), A:9 (C)
4	E	5	B:7 (D), G:8 (E), A:9 (C)
5	B	7	G:8 (E), A:9 (C)
6	G (goal)	8	—

Explanation: UCS expands the node with the lowest path cost first. It updates the cost of G when a cheaper path is found through E.

Justification of Corrections

BFS: Explores nodes level by level and uses a queue.

DFS: Uses a stack and explores one branch deeply before backtracking.

UCS: Expands the node with the lowest cumulative path cost and updates cheaper frontier paths.

(d) Apply the Corrected Traces

(i) Comparison Table

Strategy	Path found	Path cost	Nodes removed from frontier	Optimal here?
BFS	S–D–G	9	6	No
DFS	S–C–A–B–G	14	5	No
UCS	S–D–E–G	8	6	Yes

(ii) Why Is an Explored Set Needed in Graph Search?

An explored set prevents a node from being expanded repeatedly. For example, in BFS, node D is connected to both S and B. The explored set prevents repeated expansion and unnecessary work.

Conclusion: BFS explores level by level, DFS explores deeply, and UCS finds the lowest-cost path by expanding the cheapest available node first.

QUESTION 4: Search with Partial Information – Error Analysis of Belief-State Computation

(a) Identify Every Error

No.	Task / Step / Conclusion	What is wrong?
1	Task A – Step 1	The belief state after Suck is correct: {S5, S7}.
2	Task A – Step 2	The belief state after Right is incorrect. The correct belief state is {S6, S8}.
3	Task A – Step 3	The final belief state is incorrect because the agent must apply Suck to every possible state.
4	Conclusion C1	The student incorrectly concludes that sensorless problems cannot be solved. A plan can reach the goal from every possible state.
5	Task B – Step 0	The updated belief state after percept [Left, Dirty] is incorrect. It should contain only states where the agent is on the left and the left square is dirty.
6	Task B – Step 1	The predicted belief state after Suck and the updated belief state after [Left, Clean] are incorrect.
7	Task B – Step 2	The predicted belief state after Right and the updated belief state after [Right, Dirty] are incorrect.
8	Conclusion C2	The student incorrectly concludes that the goal has been reached. A dirty square remains in the world.
9	Conclusion C3	The student incorrectly states that there are only eight belief states. With eight physical states, there are 256 possible belief states.

(b) Root Cause Analysis

No.	Root cause
1	The student misunderstood how actions transform every state in a belief state.
2	The student did not correctly apply the Right action to all possible physical states.
3	The student did not update the belief state according to the effect of Suck.
4	The student confused uncertainty about the current state with the inability to reach a goal.
5	The student did not correctly filter states using the received percept.
6	The student misunderstood how actions and percepts update belief states.
7	The student did not correctly predict the agent's location and dirt status after moving right.
8	The student confused knowing the current state with achieving the goal.
9	The student confused physical states with belief states.

(c) Corrected Tables and Conclusions

(i) Task A – Corrected Belief States

Initial belief state: {S1, S3, S5, S7}

Step	Action	Belief state after action
0	Start	{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}
3	Suck	{S6, S8}

Explanation: After Suck, the left square becomes clean. After Right, the agent moves to the right square. The final Suck cleans the right square in states where it is dirty.

(ii) Task B – Corrected Belief States

Step	Action	After action (predicted)	Percept	After percept (updated)
0	Start	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S3}
1	Suck	{S5, S7}	[Left, Clean]	{S5, S7}
2	Right	{S6, S8}	[Right, Dirty]	{S6}

(iii) Corrected Conclusions C1 and C2

C1: Sensorless problems can be solved if a sequence of actions guarantees the goal from every possible initial state. The agent does not need to know its exact physical state.

C2: The final belief state is {S6}. In state S6, the agent is on the right square, the left square is clean, and the right square is dirty. Therefore, the goal has not been reached.

(d) Apply the Corrected Concepts

(i) How many different belief states are possible with eight physical states?

The number of possible belief states is:

$$
2^n
$$

Here, $n = 8$.

$$
2^8 = 256
$$

Answer: There are 256 possible belief states.

Why is belief-state search harder?

Belief-state search is harder because each belief state can contain different combinations of physical states. This creates a larger search space than searching over individual physical states.

(ii) Apply the plan [Right, Suck]

Initial belief state: {S1, S3, S5, S7}

Step	Action	Belief state after action
0	Start	{S1, S3, S5, S7}
1	Right	{S2, S4, S6, S8}
2	Suck	{S2, S4, S6, S8}

Is it a solution? No.

Reason: After moving right and performing Suck, the right square becomes clean, but the left square may still be dirty. Therefore, this plan does not guarantee that both squares are clean